

**ORIGINAL ARTICLE**

Aflatoxins and Ochratoxin A contamination in chicken tissues and eggs in three regions of Cameroon

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Abstract

Aflatoxins and Ochratoxin A are heat stable mycotoxins with side effects on human health once found in food. This study aimed to evaluate Total Aflatoxin (AFT), Aflatoxin B1 (AFB1) and Ochratoxin A (OTA) contents in fresh and boiled frozen broiler tissues and in layer eggs. The samples were collected in some farms in the Centre, Littoral and West regions of Cameroon and analyzed for their mycotoxin contents by Enzyme linked immunosorbent assay (ELISA). Results showed that all samples contained total aflatoxins, aflatoxin B1 and Ochratoxin A. Amongst samples from the Centre and Littoral, broiler gizzard was seen as the organ with high level total aflatoxins (4.5 ± 0.5 µg/kg for Centre and 5.0 ± 0.6 µg/kg for Littoral) and AFB1 (2.0 ± 0.03 µg/kg for Centre and 2.5 ± 0.2 µg/kg for Littoral) compared to broiler muscle and liver. Egg was also contaminated with AFT and AFB1 but in lower concentrations than other broiler tissues. In the later OTA was found at high concentrations in the gizzard with 5.0 ± 0.7 µg/kg (Centre region) and muscle with 1.5 ± 0.5 µg/kg (Littoral region) with the gizzard containing the lowest concentration in OTA (0.8 ± 0.1 µg/kg). In the West region broiler gizzard showed high concentrations in AFT (6.5 ± 1.1 µg/kg) and AFB1 (1.6 ± 0.2 µg/kg) and less in OTA (0.1 ± 0.05 µg/kg). Fresh bulk egg sample had the lowest concentrations in AFT (1.0 ± 0.3 µg/kg) and AFB1 (0.8 ± 0.1 µg/kg) but the highest concentration in OTA (1.7 ± 0.5 µg/kg) than AFT and AFB1 analyzed samples.

Practical application

Chicken meat and eggs are frequently consumed by the population in the Centre, West and Littoral regions of Cameroon, therefore is important to follow the quality of these products in order to reduce health risks related to the presence of carcinogens like Aflatoxins and Ochratoxins

Key words: Mycotoxins, contamination, ELISA, broiler tissues, layer eggs

1. Introduction

Mycotoxins are known as secondary metabolites produced as a result of fungal attack (Binder, 2007). The most of them have relatively small molecular weights, and usually heat-stable compounds. The significant mycotoxins in terms

of their toxicity and occurrence are aflatoxins, ochratoxins, trichothecenes, patulin, fumonisins and zearalenone (Herebian *et al.*, 2009; Iqbal *et al.*, 2014). To date the most toxic classes from a public health perception are, Aflatoxins (AFs), Deoxynivalenol (DON), Ergot Alkaloids,



Fumonisin (FBs), Ochratoxin A (OTA), Patulin (PAT), and Zearalenone (ZEN) (Binder *et al.*, 2007; Iqbal *et al.*, 2014).

Aflatoxins and Ochratoxin A are mainly found in food intended for animal and human consumption (Raters & Matissek, 2008; Tchana *et al.*, 2010; Adetunji *et al.*, 2014; Pouokam *et al.*, 2017). Aflatoxin B1 (AFB1) is recognized as the most acute and toxic class of AFs (Pittet, 1998). The International Agency for Research on Cancer has placed AFs in group 1 carcinogens, OTA in group 2B (IARC, 1993, 1999). It has been observed from previous studies that fungi can infect and subsequently produce mycotoxins in agricultural crops during pre-harvest or post-harvest conditions (Abramson *et al.*, 2005; Azaiez *et al.*, 2015). These mycotoxins are equally found in food of animal origin (Pouokam *et al.*, 2017). Chicken meats and eggs are sources of high quality nutrients such as proteins that play an important role in human nutrition (Bordoni & Danesi, 2017). In Cameroon, poultry industries has been estimated to share 42% of the market with a per capita consumption of 5.6 kg of chicken and 52 eggs annually (Höfler, 2018). Keutchatang *et al.* (2022) reported the presence of AFs, AFB₁ and OTA in chicken tissues, and eggs. This occurrence can be related to the contaminated animal feed which is the major cause of exposure of these mycotoxins to animals and therefore ultimately to humans (Volkel *et al.*, 2011; Bryden, 2012; Oyekemi *et al.*, 2018). Previous studies have indicated the presence of mycotoxins in poultry feed in Cameroon (Abia *et al.*, 2013a; Abia *et al.*, 2013b; Keutchatang *et al.*, 2023). Considering the above facts and very limited studies on the occurrence of these mycotoxins in chicken meat and eggs from Cameroon, the present study was designed to investigate the level of AFs, OTA in chicken meat

and eggs from farms of three regions of Cameroon.

2. Materials and Methods

2.1 Sample collection and pre treatment

Broilers and eggs were collected in 12 poultry farms in the Centre, Littoral and West regions of Cameroon. From each farm, 8 Broilers per farm and 30 eggs were collected as shown in table 1. Chicken were slaughtered and muscle, liver and gizzard were removed, weighed and divided into two parts. The first part was used immediately for aflatoxins and Ochratoxin A analysis and the second was stored at - 80°C for 4 months, and analyzed. Eggs were washed, broken, and introduced into 50 ml centrifuge and divided into two parts. The first part was subjected to the same treatment as broiler tissues. The second part of each sample (muscle, liver, gizzard and egg) stored at -80°C was removed after 4 months, boiled at 100°C±2°C for 30 minutes and used for the determination of aflatoxins and Ochratoxin A contents. All the samples were analyzed using ELISA.

Table 1: Number of samples collected in selected farms

Regions	Farms	Broilers	Eggs
Centre	4	32	120
West	4	32	120
Littoral	4	32	120
Total	12	96	360

2.2 Aflatoxins and ochratoxin A quantification

Total aflatoxin (AFT), AFB₁ and OTA contents were determined using quantitative ELISA (enzyme linked immune sorbent assay) kits for AFT, AFB1 and OTA in, poultry and eggs (CAT. N° 1030, 1034 and 1036, BIOO Scientific Corp., MaxSignal®, Burleson Road Austin, TX 78744

USA). Samples (2 g of ground bulk chicken tissue or 2 g of egg) were mixed with 25 mL of 70 % methanol (HPLC grade Merck) in 50 mL falcon tubes for 10 min on a vortex, followed by centrifugation at 4,000 g for 10 min using the Rotofix 32 A centrifuge (Germany). Then, 100µL of the obtained supernatant was diluted with 700µL of a mixture of methanol (HPLC grade, Merck) and extraction buffer solutions prepared as indicated by the kit manufacturer. The new diluted mixtures were used for total AFT, AFB₁ and OTA analyses following the kit manufacturer instructions. The AFT, AFB₁ and OTA contents were inversely proportional to the colour intensity established using an automated microplate reader (EL × 800, BIOTEK, Instruments Inc., Winooski, VT, United States) at 450 nm. AFT, AFB₁ and OTA content of each sample was calculated based on a calibration curve that was plotted. The lower and upper limits of quantification of the kit were 1 and 10 ppb (µg/kg) for AFT and AFB₁ and 0.2 and 1 ppb (µg/kg) for OTA, according to the supplier's information.

2.3 Statistical analysis

Data obtained were expressed as mean ± standard deviation. Student paired T test was used to compare mean between fresh and boiled frozen samples. The differences were considered significant at 95 % confident interval with p-value < 0.05.

3. Results

All fresh and unfrozen broiler tissues (muscle, liver and gizzard) and layer egg were contaminated by total Aflatoxins, Aflatoxin B₁ and Ochratoxin A (Table 2). Broiler gizzard was the most contaminated tissue in total aflatoxins (4.5 ± 0.5 µg/kg and 5.0 ± 0.6 µg/kg for Centre and for Littoral respectively) and AFB₁ (2.0 ± 0.03 µg/kg for Centre and 2.5 ± 0.2 µg/kg for Littoral).

Eggs even though contaminated in AFT and AFB₁ presented the lowest concentrations.

High concentrations of OTA was found in the gizzard and muscle (5.0 ± 0.7 µg/kg and 1.5 ± 0.5 µg/kg respectively) of broilers collected in the Centre but gizzard from Littoral region had the lowest concentration in OTA (0.8 ± 0.1 µg/kg). In the West region broiler gizzard showed high concentrations in AFT (6.5 ± 1.1 µg/kg) and AFB₁ (1.6 ± 0.2 µg/kg) and less concentration in OTA (0.1 ± 0.0 µg/kg) (Table 3). Fresh bulk egg sample had low concentration in AFT (1.0 ± 0.3 µg/kg) and AFB₁ (0.8 ± 0.1 µg/kg) but the highest concentration in OTA (1.7 ± 0.5 µg/kg) than AFT and AFB₁ analyzed samples.

As in fresh unfrozen samples, total aflatoxins (AFT), aflatoxin B₁ (AFB₁) and Ochratoxin A (OTA) were found in boiled frozen samples (broiler muscle; liver and gizzard and layer egg) from the three regions but at different concentrations. Table 2 shows that AFT concentrations increased by 33.3% from fresh muscle to boiled frozen muscle and by 20% from fresh layer egg to boiled frozen egg. At the same time, AFT concentrations decreased by 20 % from the fresh liver to boiled frozen liver and by 31.1% from fresh gizzard to boiled frozen gizzard. There was also a 40 % increase observed in the boiled frozen muscle compared to the fresh muscle sample and a 10 to 40% decrease in the other boiled frozen samples. An increase of OTA concentrations was observed in boiled frozen muscle, liver and gizzard from 13.3 to 75.33% compared to corresponding fresh samples, while a decrease of 82.4% of the concentrations of OTA was observed between fresh layer eggs to boiled frozen layer of eggs sample.

Table 3 shows the paired t test showed a significant difference between AFT concentrations in fresh broiler muscle and boiled frozen muscle.

Table 2: Total Aflatoxin, Aflatoxin B₁ and Ochratoxin A levels in chicken tissues and eggs collected from three regions in Cameroon

Mycotoxins	Samples	Centre		Littoral		West		Norms (µg/kg)
		Mean Concentrations (µg/kg)						
		Fresh	Boiled frozen	Fresh	Boiled frozen	Fresh	Boiled frozen	
Total Aflatoxin	Muscle	1.5 ± 0.5 ^a	2.0 ± 0.6 ^a	2.3 ± 0.8 ^a	3.5 ± 0.8 ^b	1.6 ± 0.5 ^a	2.0 ± 0.7 ^a	5-10 (CE, 2006)
	Liver	1.5 ± 0.03 ^a	1.3 ± 0.2 ^a	2.2 ± 0.7 ^b	1.6 ± 0.4 ^a	1.9 ± 0.3 ^a	1.4 ± 0.2 ^a	
	Gizzard	4.5 ± 0.5 ^a	3.1 ± 0.4 ^b	5.0 ± 0.6 ^a	3.5 ± 0.4 ^b	6.5 ± 1.1 ^c	4.5 ± 0.3 ^a	
	Egg	1.0 ± 0.02 ^a	1.2 ± 0.4 ^a	0.8 ± 0.4 ^a	0.9 ± 0.4 ^a	1.0 ± 0.3 ^a	1.2 ± 0.2 ^a	
Aflatoxin B ₁	Muscle	1.0 ± 0.07 ^a	1.4 ± 0.1 ^a	1.1 ± 0.2 ^a	1.8 ± 0.1 ^a	1.1 ± 0.2 ^a	2.8 ± 0.3 ^b	5-10 (CE, 2006)
	Liver	1.5 ± 0.5 ^a	0.9 ± 0.3 ^b	0.7 ± 0.1 ^b	0.5 ± 0.4 ^b	0.8 ± 0.3 ^b	0.5 ± 0.1 ^b	
	Gizzard	2.0 ± 0.03 ^a	1.6 ± 0.2 ^b	2.5 ± 0.2 ^a	1.8 ± 0.2 ^b	1.6 ± 0.2 ^b	1.2 ± 0.2 ^b	
	Egg	1.0 ± 0.02 ^a	0.9 ± 0.2 ^a	0.8 ± 0.1 ^a	0.7 ± 0.4 ^a	0.8 ± 0.1 ^a	0.7 ± 0.1 ^a	
Ochratoxin A	Muscle	1.5 ± 0.5 ^a	12.8 ± 6.0 ^b	1.5 ± 0.5 ^a	10.5 ± 2.3 ^b	1.2 ± 0.4 ^a	9.0 ± 3.0 ^c	5-20 (CE, 2006)
	Liver	5.0 ± 0.7 ^a	8.8 ± 1.8 ^b	0.8 ± 0.1 ^c	1.4 ± 0.9 ^c	1.0 ± 0.1 ^c	1.8 ± 0.1 ^c	
	Gizzard	1.5 ± 0.9 ^a	1.7 ± 0.8 ^a	1.1 ± 0.04 ^a	1.1 ± 0.08 ^a	1.1 ± 0.05 ^a	1.1 ± 0.01 ^a	
	Egg	1.7 ± 0.4 ^a	1.5 ± 0.01 ^a	2.2 ± 0.2 ^b	2.0 ± 0.3 ^b	1.7 ± 0.5 ^a	1.5 ± 0.06 ^a	

Data are presented as mean ± SD (a-c) Values within the same row with different superscripts are significantly (p<0.05) different

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Significant difference was also observed in AFB₁ concentrations in fresh broiler liver and boiled frozen broiler liver, same between fresh broiler

gizzard and boiled frozen gizzard, the differences were observed for fresh layer egg and boiled frozen layer eggs. The same significant difference was observed in OTA concentrations in fresh broiler liver and boiled frozen broiler liver, fresh broiler gizzard and boiled frozen gizzard. The same tendencies were observed in samples from the Littoral region where an increase of AFT concentrations by 52.2 % and 12.5 % were observed in boiled frozen samples of broiler muscle and layer egg respectively. An increase of 60.0 % and 85.5 % was also observed respectively for AFB₁ and OTA concentrations in boiled frozen broiler muscle compared to fresh broiler muscle. The paired t test showed significant differences (p<0.054) between AFT concentrations in various fresh and boiled samples of muscle, liver, gizzard and egg. With regards to AFB₁ concentrations, boiled liver, gizzard and eggs were significantly different (p<0.05) from that of fresh tissues whereas OTA concentrations in fresh broiler liver and muscle were significantly different (p<0.05) from that of frozen sample from the Littoral region. Mycotoxins concentrations increase was observed in boiler frozen muscle and liver from the West region (Table 3). The paired t test showed significant differences (p<0.05) between AFT concentration in fresh layer egg and boiled frozen egg; AFB₁ concentration in fresh broiler gizzard and boiled frozen gizzard and OTA concentration in fresh broiler muscle and boiled frozen muscle.

4. Discussion

Mycotoxins are secondary metabolites of a global concern; they have affected animal and human health for thousand years. These toxins are produced by certain moulds (fungi, mainly *Aspergillus*, *Penicillium* and *Fusarium*) that grow on a variety of different crops and foodstuffs

under certain circumstances (e.g. warm and humid conditions) during harvesting, handling, storage, and processing. The present study reveals that broiler tissues and layer eggs are susceptible to be contaminated by mycotoxins like Total Aflatoxin (AFT), Aflatoxin B₁ (AFB₁) and Ochratoxin A (OTA). Fresh broiler muscles and livers from the Littoral region were more contaminated with total aflatoxins than fresh samples from the two other regions (Centre and West). Fresh gizzard sample from the West region showed the highest concentration of AFT. Aflatoxin B₁ were found in high concentrations in fresh broiler muscle from the Littoral and West regions. AFB₁ was also found in fresh broiler liver from the Centre region and in fresh broiler gizzard from the Littoral region. Ochratoxin A was found in high concentrations in fresh broiler muscles from the Centre and Littoral regions and in fresh broiler liver and gizzard from the Centre region. This is probably because gizzard is the organ where feeds are stored and grind before their possible distribution in the animal's body (Gautier, 2001). In the Centre region fresh broiler liver was the most contaminated sample with OTA, while fresh broiler gizzard was the most contaminated sample with OTA in the Littoral and West region. These differences in mycotoxin contamination of each sample could be explained by the level of feed contamination and the metabolism of these mycotoxins once in animal bodies. Concerning the contamination of eggs, fresh layer eggs from Centre and West regions had high concentrations of AFT and AFB₁. While fresh layer egg from the Littoral had high concentration of OTA. It has been reported that after resorption, the largest quantity of OTA can be found in kidneys and liver, and to a considerable smaller concentration in the muscles (Galtier, 1991). In the present study, OTA concentration was higher in other tissues than in

the liver sample. However AFT, AFB₁ and OTA concentrations were higher in boiled frozen broiler muscle samples than in fresh broiler muscle samples from the three regions. This increase of mycotoxin concentrations could be explained by the fact that heat favors the release of the mycotoxins present in hidden parts of the boiled frozen broiler muscles which would not have been the case in the samples of broiler muscles analyzed fresh.

The difference of concentrations in mycotoxins between the studied regions can be explained by the climate differences between the regions and the storage conditions of poultry feed. In developing countries like Cameroon, the preservation of food and commodities is still a concern. Indeed, when foodstuffs are not stored in suitable conditions, especially during storage, molds can proliferate and lead to the occurrence of mycotoxins in foodstuffs. Every year a significant percentage of grains are contaminated with hazardous mycotoxins, thus these toxins are frequently detected in poultry feed mainly compose of grains (Keutchatang *et al.*, 2023). Some studies also reported that the minimum and optimum conditions for aflatoxins production are 27° C and 40° C- 42°C (Motalebi *et al.*, 2008). Preservation of food commodities in Cameroon is still of great concern in many communities. As recently highlighted by the narrative prevention project of the NGO NOODLES (Frazzoli & Mantovani, 2020) food agricultural and manufacturing practices currently adopted in Cameroon are often very suitable to mold proliferation and mycotoxins production. In Cameroon, few studies investigated the contamination of foodstuffs by mycotoxins. A large number of commodities including cereals (such as maize, sorghum, and rice) and derivatives mainly used when producing poultry feeds (Njobeh *et al.*, 2010; Djoulde, 2013; Kana *et al.*,

2013). Previous studies reported that poultry feed and their ingredients are good substrates for fungal growth, these fungi are able to secrete secondary metabolites known as mycotoxins (Oyekemi *et al.*, 2018).

consumption in Cameroon reported that the estimated daily consumption average for chicken varies from 12.5 g (children) to 85.9 g (elderly people) for chicken, while for eggs it ranges from 15.5 g (children) to 48 g (teenagers).

Table 3: Paired samples Test in the three region

Pairs	Mycotoxins	Samples	Centre		Littoral		West	
			t	p-value	t	p-value	t	p-value
Pair 1	AFT	Broiler's muscle - Boiled frozen broiler's muscle	0,00	1,000	-7,93	0,00	0,882	0,407
Pair 2		Broiler's liver - Boiled frozen broiler liver's	1,76	0,121	-4,41	0,003	-	-
Pair 3		Broiler's gizzard - Boiled frozen broiler's gizzard	52,03	0,000	-4,41	0,003		
Pair 4		Layer egg - Boiled frozen layer egg	-	-	4,96	0,002	14,02	0,000
Pair 5	AFB1	Broiler's muscle - Boiled frozen broiler's muscle	1,98	0,088	-13,22	0,000	-	-
Pair 6		Broiler liver's - Boiled frozen broiler liver's	5,29	0,001	-4,63	0,002	-	-
Pair 7		Broiler's gizzard - Boiled frozen broiler's gizzard	-7,93	0,000	18,52	0,000	8,81	0,000
Pair 8		Layer's egg - Boiled frozen layer's egg	-5,29	0,001	3,96	0,005	-3,96	0,005
Pair 9	OTA	Broiler's muscle - Boiled frozen broiler's muscle	-298,9	0,000	-211,6	0,000	-	-
Pair 10		Broiler's liver - Boiled frozen broiler's liver	-50,26	0,000	13,22	0,000	0,000	1,000

Furthermore, mycotoxins were reported to have the possibility to cross from feed to animal body and bioaccumulate (Volkel *et al.*, 2011). In the littoral and the west region of Cameroon, humid and hot conditions increase the occurrence of mycotoxins in feeds and thus in chicken tissues and eggs. A recent survey on chicken and eggs

This average consumption is higher than the per capita consumption levels of poultry and eggs reported by Teleu & Ngatchou (2006) as 10.9 g/day (4 kg/year) and 2.5 g/day (0.9 kg/year), respectively, and the 9.81 g cooked/day for poultry and 7.15 g cooked/day for eggs obtained

during the second Cameroonian Household Budget Survey (HBS/ECAM II) in 2001 (INS, 2002).

Once in the body, Aflatoxins are metabolized in the body into active metabolites which have a great toxicity. In the present work, it can be assume that, Aflatoxin B1 is metabolized into Aflatoxin M1 and Aflatoxin B2a in the liver. Some of these active metabolites (Aflatoxin B2) and Aflatoxin M1 are reduced to cyclopentanol and aflatoxinol by the NADP link enzyme system. Aflatoxin B1 and aflatoxinol are accumulated in eggs, aflatoxinol is the major metabolite in muscle and blood of chicken (Gargees & Shareef, 2009). Aflatoxins (AFTs) are extremely toxic and their side effects are carcinogenicity, mutagenicity, teratogenicity and immunosuppression (Eaton & Gallagher, 1994). Amongst Aflatoxins, Aflatoxin B1 is the most common and the most toxic. It is considered to be the most potent hepatocarcinogen in mammals and is classified as the probable group 1 carcinogen by the international Agency of Cancer Research (IARC, 1993). Aflatoxin M1 the metabolized form of Aflatoxin B1 is a monohydric derivative carcinogen for humans and is classified by IARC as group 1 (IARC, 2002).

The difference in mycotoxin concentrations observed in fresh broiler muscle samples and in stored frozen broiler muscle samples was statistically significant. An increase of OTA concentration was also observed in boiled frozen broiler liver from the Centre region. It has been reported that once inside the liver, OTA is hydrolyzed to OT α and L-phenylalanine and other metabolites that are less toxic (Gautier *et al.*, 2001). It was also reported that primary and secondary effects of OTA are probably connected with the influence on enzymes participating in the phenylalanine metabolism and the functions of

mitochondria in one way, and the increase of lipid peroxidation in the liver and kidney microsomes in the other way (Fuchs & Hult, 1992). Known for its nephrotoxicity, carcinogenic, immunotoxic, genotoxic and teratogenic properties in all animal species tested (Pitt *et al.*, 2013), OTA is classified by the International Agency of Cancer Research in group 2B as a carcinogenic compound to human (IARC, 1999). The significant differences in mycotoxin concentrations observed could be as a result of their variation; this is highlighted by the increase of mycotoxin concentrations in some boiled frozen samples. This increase in mycotoxin concentrations in boiled frozen samples compared to the same fresh samples can be explain by the fact that the heat would have the capacity to weaken the cell membranes and to release the compounds that the freezing alone would not be enough to release.

Mycotoxins also have negative effect on chicken production and efficiency by reducing growth rate, impairing immunity and sometimes causes mortality of chicken causing economic losses to farmers. In order to reduce the occurrence, Chicken farmers in the studied regions need to ensure that the feeds they purchased are stored in proper conditions; furthermore they should avoid the usage of feeds that appear discolored, lumped together and smelling musty.

5. Conclusion

The present study investigates on the level of AFs, OTA in chicken meat and eggs from farms of three regions of Cameroon. Chicken meat and eggs increasingly consumed by the Cameroonian population are contaminated by mycotoxins like Total Aflatoxin (AFT), Aflatoxin B₁ (AFB₁) and Ochratoxin A (OTA). The population of the three regions, Centre, Littoral and West are at risk for chronic exposure due to dangerous levels

of Aflatoxins found in these products. This can pose a serious health concerns, it is therefore necessary for national authorities as government to implement control measures that are critical initial steps towards food safety, economic sustainability and public health promotion in the three region and in Cameroon. Further studies have to focus on epidemiology and toxic effects.

Conflict of interest

The authors declare that they have no competing interests.

Ethics

This Study does not involve Human or Animal Testing.

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